FACEPAY: AN AI BASED SECURE AND CONTACTLESS PAYMENT SYSTEM USING FACIAL RECOGNITION AUTHENTICATION

A Mini Project Report

submitted in partial fulfilment of the requirements for the award of the degree of

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in

ARTIFICIAL INTELLIGENCE

by

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DECLARATION

We hereby declare that the Mini Project report entitled "**FACEPAY: AN AI BASED SECURE AND CONTACTLESS PAYMENT SYSTEM USING FACIAL RECOGNITION AUTHENTICATION**" submitted to the Department of Artificial Intelligence, Anurag University, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is a record of original work carried out by us under the guidance of Mr. G. Victor Daniel, Assistant Professor, Department of Artificial Intelligence. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma. Wherever the work of others, including pretrained models, open source libraries and published papers, has been used, it has been duly acknowledged and cited.

Use of AI tools: We declare that we have used large language model tools for language editing, code debugging assistance and drafting support while preparing this report. The system design, implementation, testing and interpretation of results are our own, and we take full responsibility for the content of this report.

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CERTIFICATE

This is to certify that the Mini Project report entitled "**FACEPAY: AN AI BASED SECURE AND CONTACTLESS PAYMENT SYSTEM USING FACIAL RECOGNITION AUTHENTICATION**" is a bonafide record of the work carried out by Pallavi Mirayala (23EG106C36), Shankure Sidha Reddy (23EG106C59), Sikha Vivek (23EG106C60) and Solleti Dheeraj Reddy (23EG106C61) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence during the academic year 2026 to 2027. The work embodied in this report has not been submitted to any other university or institute for the award of any degree or diploma.

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ABSTRACT

Digital payments have become a normal part of daily life in India, from paying at a shop to sending money to family. Yet most payment apps still confirm the identity of the user with passwords, Personal Identification Numbers (PINs) and One Time Passwords (OTPs). These credentials can be forgotten, stolen, watched over the shoulder or intercepted through phishing and SIM swap fraud, and they add extra steps and waiting time that are especially hard for elderly and non technical users.

This project presents FacePay, an AI based secure and contactless payment system that uses facial recognition instead of passwords, PINs and OTPs. A user registers once with their face and account details. At the time of payment, the system captures a live face image through the device camera, detects the face, aligns it using 68 facial landmarks and converts it into a 128 value face descriptor, which is compared with the descriptor stored at registration. If the Euclidean distance between the two is below a threshold of 0.55, the user is authenticated and can complete a wallet transfer or a bank transfer.

The system is built as a three tier web application. The client tier uses HTML, CSS and JavaScript with the face-api.js library, which runs an SSD MobileNet v1 face detector with a minimum confidence of 0.5, a 68 point landmark model and a ResNet style face recognition network inside the browser. The application tier uses Node.js for transaction logic, and Firebase Realtime Database stores user profiles, face descriptors (never raw photographs) and an auditable transaction log in which every attempt, accepted or rejected, is recorded.

The report reviews earlier work on face embeddings, face detection, face based payment systems and face anti spoofing, and identifies the gap that FacePay fills: most existing work either studies face recognition alone or links it to only one payment channel or to extra hardware. FacePay combines face verification, wallet and bank transfers and secure cloud storage in one continuous flow. The report defines an evaluation framework based on accuracy, False Acceptance Rate, False Rejection Rate, authentication time and user convenience, presents functional test cases, and openly states the main limitation of the current version, which is the absence of liveness detection. Liveness detection, fraud detection and QR and UPI integration are planned as future work.

Keywords: facial recognition; biometric authentication; contactless payment; face embedding; digital wallet; Firebase; secure transactions

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
FAR	False Acceptance Rate
FRR	False Rejection Rate
HTML	HyperText Markup Language
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
LBP	Local Binary Pattern
LFW	Labeled Faces in the Wild
MTCNN	Multi task Cascaded Convolutional Network
OTP	One Time Password
PAD	Presentation Attack Detection
PIN	Personal Identification Number
QR	Quick Response
REST	Representational State Transfer
rPPG	Remote Photoplethysmography
SDK	Software Development Kit
SIM	Subscriber Identity Module
SSD	Single Shot MultiBox Detector
TLS	Transport Layer Security
UI	User Interface
UML	Unified Modeling Language
UPI	Unified Payments Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Digital payments have become a normal part of everyday life. People now pay at grocery shops, book train and movie tickets, pay electricity bills, send money to family members and buy things online through mobile wallets, UPI apps and net banking. In India especially, the growth of digital payments in the last few years has been very fast, and even small street vendors now display a QR code for payments. As more and more money moves through these systems, one question has become very important: how does the payment system confirm that the person making the payment is really the owner of the account?

Most payment platforms today answer this question using passwords, Personal Identification Numbers (PINs) and One Time Passwords (OTPs). These methods have been in use for many years and people are familiar with them. However, each of them has known weaknesses, and attackers have become very good at exploiting those weaknesses. At the same time, biometric technology, and face recognition in particular, has improved a lot because of deep learning [1], [2], [3]. This project, FacePay, brings these two areas together and uses the face of the user as the key to authorise a payment.

1.1.1 Overview of FacePay

FacePay is an AI based secure and contactless payment system that uses facial recognition instead of passwords, PINs and OTPs. A user registers once by entering basic details and allowing the camera to capture their face. The system converts the face into a list of 128 numbers called a face descriptor and stores only this descriptor, not the photograph. Later, when the user wants to make a payment, the system captures a live image of the face through the camera of the laptop or phone, computes a new descriptor and compares it with the stored one. If the two descriptors are close enough, the user is authenticated and is allowed to complete a wallet transfer or a bank transfer.

The system is built as a web application with three tiers. The front end is written in HTML, CSS and JavaScript and runs the face detection and recognition models inside